
Data 250 Practice Exam 3

1.

$$2x + 3y + 5z = 8$$

$$x - y + z = 0$$

- a. Write this system of equations as a matrix equation.
b. Solve the system of equations for the general solution.

2. Given three $n \times n$ matrices $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$

Write the i -th row and j -th column of the matrix multiplication ABC using summation notation.

3. Solve for the matrix X given that $AXA^{-1} = Y$

4. Given that the 6×7 matrix X has its null space spanned by $[1, 3, -1, 0, 0, 2, 5]$ a) what is the rank of X ?

b) what is the dimension of the image of $T(v) = Xv$?

c) Let $Y = XX^t$. Show that Y is a symmetric matrix.

d) Show that Y is invertible.

5. $A = \begin{pmatrix} 2 & 3 \\ x & 6 \end{pmatrix}$

a) For what values of x is A nonsingular?

b) Calculate A^{-1} when $x = 1$

6. Suppose $Av = 3v$ and $Au = 2u$ where $u = [1, 1]$ and $v = [1, 2]$

a) Let $e_1 = [1, 0]$ and $e_2 = [0, 1]$. Calculate Ae_1 and calculate Ae_2

b) Find the four entries a_{ij} of A

c) What are the eigenvalues of A ?

7. Suppose the singular values of A are 3, 2, 1. What is the best estimate for the norm of Av for a nonzero vector v ?